Kinds of convergence

We cover the concept of convergence in statistics

Noara Razzak

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1. Almost sure convergence

$$T_n \xrightarrow[n\to\infty]{a.s} T \text{ if and only if } P[\{T_n(w) \xrightarrow[n\to\infty]{} T(w)\}] = 1$$

This is the strongest form of convergence.

2. Convergence in probability

$$T_n \xrightarrow[n\to\infty]{P} T \text{ if and only if } P[|T_n(w) - T(w)|\} \geq \epsilon] \xrightarrow[n\to\infty]{} 0 \text{ , for all } \epsilon > 0$$

Convergence in probability is weaker than almost sure convergence.

3. Convergence in Quadretic Mean

$$T_n \xrightarrow[n\to\infty]{L^2} T \text{ if and only if } E[|T_n(w) - T(w)|^2] \xrightarrow[n\to\infty]{} 0$$

Convergence in quadratic mean is slightly weaker than almost sure convergence but stronger than convergence in probability.

4. Convergence in Distribution

$$T_n \xrightarrow[n\to\infty]{(d)} T \text{ if and only if } P[T_n(w) \leq x] \xrightarrow[n\to\infty]{} P[T(w) \leq x]$$

Convergence in distribution is the weakest form of convergence.

The following relationships holds:

(a)

$$X_n \xrightarrow{a.s} X \text{ implies that } X_n \xrightarrow{P} X$$

If X_n converges almost surely to X then it must converge in probability. The reverse does not hold.

(b)

$$X_n \xrightarrow{L^2} X \text{ implies that } X_n \xrightarrow{P} X$$

If X_n converges in quadratic mean to X then it must converge in probability. The reverse does not hold.

(c)

$$X_n \xrightarrow{P} X \text{ implies that } X_n \xrightarrow{(d)} X$$

If X_n converges in probability to X then it must converge in distribution. The reverse does not hold.

Theorem: Continuous Mapping Theorem

Let X_n, X, Y_n, Y be random variables. Let g be a continuous function.

(a) If $X_n \xrightarrow{P} X$, then $g(X_n) \xrightarrow{P} g(X)$

(b) If $X_n \xrightarrow{(d)} X$, then $g(X_n) \xrightarrow{(d)} g(X)$

Examples

Example 1:

Suppose $X_n \xrightarrow{P} \mu$. If we define $g(X_n)$ as nX_n then $g(X_n) \xrightarrow{P} n\mu$

Example 2:

Suppose $X_n \sim N(0, 1/n)$. If we define $g(X_n)$ as $\sqrt{n}X_n$ then $g(X_n) \sim N(0, 1)$.

Theorem: The Law of Large Numbers (WLLN)

If $X_1, ..., X_n$ are iid, then $\bar{X}_n \xrightarrow{P} \mu$, where $\bar{X}_n = n^-1 \sum_{i=1}^{n} X_i$. This simply means that sample mean converges to population mean as n becomes large.

Theorem: Central Limit Theorem

Let $X_1, ..., X_n$ are *iid* with mean μ and variance σ^2. Let $\bar{X}_n = n^-1 \sum_{i=1}^{n} X_i$. Then:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0, 1)$$

The Central Limit Theorem implies that the sample mean is always distributed normally regardless of the original distribution.